PX-C single-file continuous physical runtime - exhaustive active specification v1

Accepted PXR0 v2 parent b76f4f1c276555a6f80ff697dbc9b9ef850df76e; PX-C candidate
Canonical source SHA-256: e34a9442205fa63d4bde3d286fb7c0c6e722ba04b64c403535e1db71cf3fb8aa
Retained state: cells, arrows, queued spikes, local tick, generation/live state, resistance, eligibility and activation.
Local rules: threshold/refractory firing, Drive traversal, qualified Modulation, decay, pressure, deallocation and nearby proposal.
Continuous loop: arrive appends anonymous physical inputs and advances the retained queue to natural quiescence; later calls retain the same body.
External crossings: arrive returns only ordinary crossings into its requested outward region; RunResult also reports bounded work, quiescence and resident bytes.

All active types/state (13)

CellId - tuple index naming one cell.
ArrowId - tuple index naming one arrow.
CellSpec - physical_id, position, region, threshold, resistance bootstrap fields.
TransmissionMode - Drive or Modulatory physical transmission.
ArrowSpec - from, to, delay, phase, coupling, resistance, mode bootstrap fields.
SpikeInput - arrival_tick, phase, origin_physical, target, impulse boundary fields.
Cell - physical_id, position, region, threshold, activation, last tick, refractory limit, generation, resistance, live.
Arrow - from/to, delay, phase, coupling, source generation, generation, resistance, live, eligibility, mode.
Spike - arrival tick, phase, physical origin, target/generation, impulse, serial, optional arrow/generation.
Crossing - tick, physical endpoints, endpoint regions and impulse.
Work - total plus Drive, Modulation, local-update, proposal and deallocation counts.
RunResult - crossings, work, natural-quiescence flag and resident bytes.
PlasticSubstrate - cell/arrow/spike vectors, tick, next serial and pressure tick.

All active functions/methods (16)

Work::total - returns accumulated bounded-work count without mutation.
PlasticSubstrate::new - creates empty physical state.
PlasticSubstrate::add_cell - validates and appends one generic cell.
PlasticSubstrate::add_arrow - validates endpoints/delay and appends one generic arrow.
PlasticSubstrate::enter - validates time/target and appends one external spike.
PlasticSubstrate::arrive - enters anonymous inputs, propagates to quiescence, then selects outward-region crossings.
PlasticSubstrate::advance_time - applies local elapsed-time dynamics with an empty queue.
PlasticSubstrate::propagate - delivers the ordered queue to natural quiescence and returns crossings/work.
PlasticSubstrate::apply_modulatory_return - strengthens only live eligible arrows from the reached cell.
PlasticSubstrate::elapse_to - applies ordinary pressure, expiry pressure and cell decay.
PlasticSubstrate::propose_local_arrows - adds deterministic nearby Drive arrows after external firing.
PlasticSubstrate::decay_cell - moves activation toward zero over elapsed local time.
PlasticSubstrate::require_cell - enforces substrate-local cell identity.
PlasticSubstrate::spike_order - orders arrival tick, phase, origin, target identity and serial.
PlasticSubstrate::resident_bytes - reports cell plus arrow storage for the registered bound.
pressure_arrow - reduces resistance and retires a zero-resistance arrow by generation.

29/29 entries; one production file; no hidden module or auxiliary surface.